HIRAK MONDAL

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+8801400705050 — hirakmondal.cse@gmail.com | hirak-mondal-94b651213 |
Hirak46 | 0009-0006-4756-5529 | Hirak-Mondal-2 | Hirak Mondal
Location: Batiaghata-9260, Khulna, Bangladesh

BIOGRAPHY

Hirak Mondal (NID: 1031418609) is currently pursuing M.Sc. in Computer Science and Engineering at Khulna University, Bangladesh. He obtained a B.Sc. in Computer Science and Engineering from North Western University, Khulna. His research focuses on health informatics, neuroscience, cybersecurity, machine learning, deep learning, and artificial intelligence. He has authored and co-authored about 11 publications, including journal articles, conference papers, and book chapters.

EDUCATION

Master of Science | ***Computer Science & Engineering Discipline*** January 2025 – On Going
Khulna University, Khulna - 9208, Bangladesh.
Result: CGPA – On Going
Courses: Research Methodology & Ethics, Network Optimization, Human-Computer Interaction, Advanced Probability & Statistics, Machine Learning.

Bachelor of Science | ***Department of Computer Science & Engineering*** January 2020 – May 2024
North Western University, Khulna - 9100, Bangladesh.
Result: CGPA – 3.67 (Out of 4.00)
Major Courses: Data Structures, Algorithms, Operating Systems, Software Engineering, Computer Architecture, Computer Networks, Machine Learning, Artificial Intelligence.

WORK EXPERIENCE

Research Assistant March 2022 - December 2024
VRD Research Lab Khulna, Bangladesh.

- I support innovative research in a variety of sectors as a research assistant at VRD Research Lab. I help with research paper writing in addition to collecting and analyzing data.
- In order to accomplish research objectives, I work with team members and contribute to the design of experiments.
- My role also involves staying updated with the latest developments in the field to enhance the quality and impact of our research.
- Mail: **hirak@vrdlab.com**

PUBLICATION

Summary of Publications

Total articles - 11 (First/Corresponding author's articles marked as †), Google Scholar from here.

Journal Articles

- **Impact Factor: 3.6:** Anindya Nag, **Hirak Mondal**, Md Mehedi Hassan, Taher Al-Shehari, Mohammed Kadrie, Muna Al-Razgan, Taha Alfakih, Sujit Biswas, Anupam Kumar Bairagi. "*TumorGANet:* A Transfer Learning and Generative Adversarial Network-based Data Augmentation Model for Brain Tumor Classification." *IEEE Access* 12, (2024).

Conference Proceedings

2025

- **Conference Country: Chittagong, Bangladesh: Hirak Mondal**, Kamrul Golder, Towfeka Benta Towhid, Bushra Jamil,Anindya Nag, Anupam Tarafder, Anindya Mozumder, Md Asaduzzaman. "Enhanced Interpretability in Depression Classification with the Use of Machine Learning Models with Explainable AI." *In Proceedings of International Conference on Electrical, Computer and Communication Engineering* vol 1, May 30, (2025).†
- **Conference Country: Cox's Bazar, Bangladesh:** Kamrul Golder, Anindya Nag, Fahim Faisal Tarun, **Hirak Mondal**, Md. Mehedi Hassan, Asif Karim."*EmotionRF:* Enhancing Emotion Classification from EEG Signals Using Machine Learning with Explainable AI." *In Proceedings of 27th International Conference on Computer and Information Technology* vol 1, June 10, (2025).

2024

- **Conference Country: Bangladesh:** Anindya Nag, **Hirak Mondal**, Sheikh Ridwan Raihan Kabir, Md. Rabiul Islam, Md. Estiak Ahmed, S M Hasan Jamil."Emotion Decoding: An Extensive Examination of Electroencephalogram Signals Using Explainable Machine Learning." *In Proceedings of 6th International Conference on Electrical Engineering and Information & Communication Technology* pp. 604-609, May 23, (2024).
- **Conference Country: Bangladesh:** Sheikh Ridwan Raihan Kabir, **Hirak Mondal**, Anindya Nag, S. M. Hasan Jamil, Piya Das. *Identification of Lung Cancer Using Particle Swarm Optimization and Machine Learning Technique.* In: Mahmud, M., Kaiser, M.S., Bandyopadhyay, A., Ray, K., Al Mamun, S. (eds) *Proceedings of Trends in Electronics and Health Informatics. TEHI 2023. Lecture Notes in Networks and Systems, vol 1034,pp 421–430.* Springer, Singapore (2025). https://doi.org/10.1007/978-981-97-3937-0_29 †

Book Chapters

2025

- **Hirak Mondal**, Anindya Nag, Md. Mushfiqur Rahman, Anupam Kumar Bairagi."Interpreting Electroencephalogram Brain Signals: Insights from AI-driven Analysis."*Brain Networks in Neuroscience: Personalization Unveiled Via Artificial Intelligence, River Publishers*, (2025).†
- Tamanna Haque Ritu, **Hirak Mondal**, Anindya Nag, Riya Sil. "Brain Network Dynamics: Implications for Health and Disease."*Brain Networks in Neuroscience: Personalization Unveiled Via Artificial Intelligence, River Publishers*, (2025).

2024

- **Hirak Mondal**, Saima Siddika,Anindya Nag, and Anupam Kumar Bairagi."Diabalance ML: Enhancing diabetes detection performance through dataset balancing strategies." *Industry 6.0: Technology, Practices, Challenges, and Applications, CRC Press*, (2024).†
- **Hirak Mondal**, Saima Siddika,Anindya Nag, Riya Sil."Unveiling the Essence of Digital Twins: A Comprehensive Study on Digital Twins for Future Innovation." *Exploring the Advancements and Future Directions of Digital Twins in Healthcare 6.0, IGI Global*, (2024).†
- **Hirak Mondal**, Md. Mehedi Hassan, Anindya Nag, and Anupam Kumar Bairagi "The Integration of Federated Deep Learning with Internet of Things in Healthcare."*Federated Deep Learning for Healthcare: A Practical Guide with Challenges and Opportunities, CRC Press*, (2024).†
- Anindya Nag, **Hirak Mondal**, Md. Mehedi Hassan, Prianka Saha."Transcending Boundaries: Assessing Transfer Learning's Effectiveness in ECG-Based Heart Disease Prediction."*Soft Computing in Industry 5.0 for Sustainability, Springer, Cham*, (2024).

Membership of Scientific Societies

Institute of Electrical and Electronics Engineers (IEEE) (From 2025 - Now)
Graduate Student Member & IEEE Young Professionals

Honors and Awards

Top 10 Student Paper Awards (Paper ID: 304) 6 may 2024
6th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT)

Skills

Languages: English (Intermediate), Bangla (Native)
Programming: Python (NumPy, SciPy, Matplotlib, Pandas), C, C++, Java, C##, HTML, CSS
Platformsx: Visual Studio, Google Collab, Anaconda, PyCharm, Web, Windows, Arduino
Tools: MySQL Workbench
Document Creation: Microsoft Office Suite, LaTex
Soft Skills: Teamwork, Leadership, Communication, Project Management, Writing, Time Management

Certifications

- **ICPC 2021:** Participation in Asia Dhaka Regional Preliminary Contest.
- **ICPC 2022:** Participation in Asia Dhaka Regional Preliminary Contest.
- **Khulna University Datathon:** Participation in Final Round.
- **Intra CSE Programming Contest:** Participation at North Western University.

Volunteer Experience

- **NWU CSE FEST-2023:** Excellence as an Organizing Committee Member.
- **Volunteer ICPC 2023:** Participation as a volunteer.
- **NWU Comptron Club:** Publication Secretary.

References

Dr. Anupam Kumar Bairagi
Professor,
Computer Science and Engineering, Khulna University, Khulna -9208, Bangladesh
Email: anupam@cse.ku.ac.bd

Declaration

I hereby declare that all the details mentioned above are by the truth and fact as per my knowledge and I hold the responsibility for the correctness of the above-mentioned particulars.

Hirak Mondal